Outcome-Frame Monodromy and Exact Limits of Marginal-Only Quantum Feedback

Zenodo Archival Preprint v1.0

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September 11, 2026

Abstract

A quantum instrument is not determined by its induced positive-operator-valued measure (POVM) and nonselective channel, yet the global control consequences of this residual branch structure remain incompletely characterized. We study the nondegenerate fiber of instruments with both marginals fixed. The subgroup of outcome permutations preserving the effect tuple acts as the deck group of the ordered outcome-frame covering, and closed loops define a conjugacy-class-valued monodromy representation. Nontrivial monodromy obstructs any continuous global branch labeling based only on local process continuity. We construct an explicit cyclic family with a uniform m-outcome POVM and a fixed completely dephasing channel, derive robust overlap-based reconstruction and a conservative finite-sample guarantee, and certify entangled conditional Choi states although the nonselective channel is entanglement breaking. For a noisy version of the cyclic family, we prove an exact minimax separation in conditional feedback. Every recovery rule restricted to the two marginals and the reported local outcome has optimal worst-sector entanglement fidelity exactly $1/m$, whereas a frame-aware controller achieves the optimal value $1-\lambda+\lambda/m$. The exact operational value of the missing outcome-frame information in this model is therefore $(1-\lambda)(1-1/m)$. The results lie entirely within standard quantum mechanics and should be read as an instrument-level, permutation-valued specialization of established Kraus-frame holonomy with a distinct global-labeling and feedback limitation.

1 Introduction

A quantum instrument resolves a physical process into outcome-conditioned completely positive maps. It therefore contains information that is absent from either of its two standard marginals alone: the POVM, which fixes outcome probabilities, and the nonselective channel, which fixes average dynamics after the classical outcome has been discarded [1, 2, 3]. This distinction is operationally relevant in adaptive and dynamic circuits, where a recorded outcome controls a subsequent operation.

The nonuniqueness of instruments with common POVMs or channels is well known, and the structure, compatibility, programmability, and sequential realization of quantum instruments are active topics [4, 5, 6, 7, 8, 9, 10]. Separately, holonomy for quantum channels has been formulated through transport of Kraus frames [11], and environment-assisted error correction exploits outcome-resolved random-unitary decompositions [12, 13]. Our goal is therefore deliberately narrower than claiming a new form of channel holonomy or a new ontology.

We ask whether a family of instruments can remain fixed in both standard marginals while the recorded branch frame acquires nontrivial global transport, and what operational limitation remains if a controller is restricted to those marginals. We establish six results:

1. a fixed-marginal outcome-frame covering with deck group given by the effect stabilizer G_E;
2. a monodromy representation $\mu : \pi_1(B, b_0) \to G_E$ and the associated obstruction to continuous global branch labeling;
3. an explicit cyclic family with uniform POVM and fixed complete dephasing channel;
4. a deterministic assignment-gap robustness theorem and a conservative finite-sample reconstruction bound;
5. a Choi-entanglement certificate showing genuinely quantum conditional branches even when the common nonselective channel is entanglement breaking;
6. an exact feedback separation: marginal-only recovery has optimal worst-sector entanglement fidelity $1/m$, whereas frame-aware recovery achieves $1 - \lambda + \lambda/m$.

The last result is the main strengthening relative to the earlier manuscript version. It replaces a single explicit mismatch example by an exact minimax statement over all marginal-only completely positive trace-preserving (CPTP) recovery maps.

2 Fixed-marginal instrument fibers

Let $\mathcal{H}$ be finite-dimensional. An m-outcome quantum instrument is a tuple

$$\mathcal{J} = (\mathcal{J}_0, \ldots, \mathcal{J}_{m-1}) \tag{1}$$

of completely positive, trace-nonincreasing maps whose sum $\Phi = \sum_a \mathcal{J}_a$ is trace preserving. Its POVM effects are $E_a = \mathcal{J}_a^*(I)$.

For a labeled effect tuple $E = (E_0, \ldots, E_{m-1})$ and a channel Φ, define

$$\mathcal{F}_{\mathrm{ord}}(E, \Phi) = \left\{ \mathcal{J} : \mathcal{J}_a^*(I) = E_a,\ \sum_a \mathcal{J}_a = \Phi \right\}. \tag{2}$$

Only outcome permutations preserving the effect tuple act within this set, so define

$$G_E = \{\sigma \in S_m : E_{\sigma(a)} = E_a \text{ for all } a\}. \tag{3}$$

For the uniform POVM $E_a = I/m$, $G_E = S_m$.

Let $\mathcal{F}_{\mathrm{nd}} \subset \mathcal{F}_{\mathrm{ord}}$ be the subset in which all branch maps are pairwise distinct. The group G_E acts by relabeling,

$$(\sigma \cdot \mathcal{J})_a = \mathcal{J}_{\sigma^{-1}(a)}. \tag{4}$$

Theorem 1 (Fixed-marginal outcome-frame covering)**.** *In finite dimension, the action of G_E on $\mathcal{F}_{\mathrm{nd}}$ is free and the quotient map*

$$p : \mathcal{F}_{\mathrm{nd}} \longrightarrow \mathcal{F}_{\mathrm{nd}}/G_E \tag{5}$$

is a regular covering with deck group G_E. If a loop γ in the quotient is lifted from an ordered initial frame, then

$$\widetilde{\gamma}(1) = H_\gamma \cdot \widetilde{\gamma}(0), \qquad H_\gamma \in G_E. \tag{6}$$

Changing the initial outcome convention by $\tau \in G_E$ sends H_γ to $\tau H_\gamma \tau^{-1}$. Hence only the conjugacy class $[H_\gamma]$ is frame-independent.

The proof is elementary: pairwise distinctness makes the finite-group action free; a finite free action on a metric space is properly discontinuous; path lifting then determines the endpoint permutation. Full details are in the Supplemental Material.

Corollary 1 (Global labeling obstruction)**.** *Let B be a path-connected, locally path-connected subset of the nondegenerate unordered fixed-marginal space. The following are equivalent: (i) the restricted cover over B admits a continuous global section; (ii) the monodromy representation $\mu : \pi_1(B, b_0) \to G_E$ is trivial; and (iii) the restricted cover is globally trivial. Therefore, if $\mu([\gamma]) \neq e$, then γ cannot be contracted within the nondegenerate fixed-marginal region. Any contraction must cross a branch-degeneracy locus or leave that region.*

This does not deny the existence of externally trusted hardware labels. It identifies the obstruction faced by a labeling rule that transports branch identity solely through local continuity of the conditional processes.

3 Explicit cyclic family

Set the system dimension equal to the number of outcomes, $d = m$. In the computational basis define

$$N = \sum_{j=0}^{m-1} j|j\rangle\langle j|, \quad Z = e^{2\pi i N/m}, \quad U(\theta) = e^{i\theta N}. \tag{7}$$

Consider

$$\mathcal{J}_{a|\theta}(\rho) = \frac{1}{m} U(\theta) Z^a \rho Z^{-a} U(\theta)^\dagger. \tag{8}$$

Every branch has effect I/m, and

$$\sum_a \mathcal{J}_{a|\theta}(\rho) = \Delta(\rho) = \frac{1}{m} \sum_a Z^a \rho Z^{-a}, \tag{9}$$

where Δ is complete dephasing in the computational basis. Because

$$U(\theta + 2\pi/m) = U(\theta) Z, \tag{10}$$

we have

$$\mathcal{J}_{a|\theta+2\pi/m} = \mathcal{J}_{a+1 \,(\mathrm{mod}\, m)|\theta}. \tag{11}$$

Thus $\theta \in [0, 2\pi/m]$ is a closed loop in the unlabeled quotient and induces the m-cycle $(0\,1 \cdots m{-}1)$ on the ordered frame.

For q equally spaced samples, the lifted labels $(a, k) \in \mathbb{Z}_m \times \mathbb{Z}_q$ evolve by

$$g(a, k) = \begin{cases} (a, k+1), & k < q-1, \\ (a+1, 0), & k = q-1, \end{cases} \tag{12}$$

which is a single cycle of order mq. For $m = 2, q = 4$, the lift has eight distinct labeled branch-position records; this is a property of the explicit discretized construction, not a claim of eight fundamental physical states.

A numerical audit for $m = 2, 3, 4$ and $q = 4$ reproduces the expected endpoint cycles, fixed total channel to floating-point precision, and distinct branch channels. These calculations are included in the reproducibility archive.

4 Robust reconstruction

Let $\omega_{a,k}$ be the normalized Choi state of branch (a, k). For adjacent sample points define an overlap matrix $F_k(a, b)$ and, for a permutation $\sigma \in S_m$, the assignment score

$$S_k(\sigma) = \sum_{a=0}^{m-1} F_k(a, \sigma(a)). \tag{13}$$

Let δ_k be the exact gap between the best and second-best assignment scores.

Theorem 2 (Deterministic assignment stability)**.** *If every estimated overlap differs from its exact value by at most ϵ and*

$$\delta_k > 2m\epsilon \tag{14}$$

for every loop edge, then every reconstructed edge permutation, and hence the loop monodromy, is unchanged.

The proof follows because an assignment contains m entries, so its score changes by at most $m\epsilon$ and the difference of two assignment scores by at most $2m\epsilon$.

If N overlap quantities are estimated by independent bounded Bernoulli tests, Hoeffding's inequality and a union bound give the conservative sufficient condition

$$n \geq \frac{\ln(2N/\alpha)}{2\epsilon^2} \tag{15}$$

shots per overlap for simultaneous error at most ϵ with confidence at least $1-\alpha$.

Using the audit's factor-of-two safety margin at $q = 4$ gives the budgets in Table 1.

Table 1: Conservative 95% simultaneous-confidence budgets for the explicit $q = 4$ construction. These are sufficient, not optimized lower bounds.

m	N	$\delta_{\min}$	shots/overlap	total shots
2	20	1.4142	107	2140
3	45	1.4880	244	10980
4	80	1.4957	462	36960

5 Quantum branch certification

Permutation monodromy by itself is not uniquely quantum: a classical hidden label can also be permuted. We therefore add branch-level Choi entanglement certification. Consider normalized noisy conditional channels

$$\Lambda^{(\lambda)}_{a|\theta} = (1-\lambda)\,\mathrm{Ad}_{U(\theta)Z^a} + \lambda\Delta, \tag{16}$$

with instrument branches $\mathcal{J}^{(\lambda)}_{a|\theta} = \Lambda^{(\lambda)}_{a|\theta}/m$. The POVM and nonselective channel remain fixed.

Let $\Omega^{(\lambda)}_{a,\theta}$ be the normalized Choi state and $|U_{a,\theta}\rangle$ the ideal maximally entangled Choi vector. Direct partial-transpose block diagonalization gives

$$\mathcal{N}\left(\Omega^{(\lambda)}_{a,\theta}\right) = \frac{(m-1)(1-\lambda)}{2}, \tag{17}$$

and

$$\langle U_{a,\theta}|\Omega^{(\lambda)}_{a,\theta}|U_{a,\theta}\rangle = 1 - \lambda\left(1 - \frac{1}{m}\right). \tag{18}$$

Any separable Choi state has overlap at most $1/m$ with a maximally entangled state. Hence every $\lambda < 1$ excludes measure-and-prepare conditional branches, although the common nonselective channel Δ is entanglement breaking.

6 Exact limits of marginal-only feedback

We now make the outcome-frame ambiguity operational. Let the normalized conditional channels be

$$\Lambda^{(\lambda)}_{a,r} = (1-\lambda)\,\mathrm{Ad}_{UZ^{a+r}} + \lambda\Delta, \qquad r \in \mathbb{Z}_m, \tag{19}$$

where the sector r specifies the global outcome-frame offset. A marginal-only recovery rule is a family of CPTP maps $\{\mathcal{R}_a\}$ that may depend on the reported local outcome a but not on r. Define

$$\mathcal{C}_r = \frac{1}{m}\sum_a \mathcal{R}_a \circ \Lambda_{a,r}^{(\lambda)}. \tag{20}$$

Theorem 3 (Exact marginal-only minimax fidelity)**.** *For the family above,*

$$\sup_{\{\mathcal{R}_a\}} \min_{r\in\mathbb{Z}_m} F_{\mathrm{e}}(\mathcal{C}_r, \mathrm{id}) = \frac{1}{m}. \tag{21}$$

Proof. For fixed a, averaging over r gives $m^{-1}\sum_r \Lambda_{a,r}^{(\lambda)} = \Delta$. Hence

$$\frac{1}{m}\sum_r \mathcal{C}_r = \frac{1}{m}\sum_a \mathcal{R}_a \circ \Delta. \tag{22}$$

Each $\mathcal{R}_a \circ \Delta$ is entanglement breaking, as is their convex mixture, so its normalized Choi state overlaps the maximally entangled target by at most $1/m$. Since entanglement fidelity is linear in the channel,

$$\frac{1}{m}\sum_r F_{\mathrm{e}}(\mathcal{C}_r, \mathrm{id}) \le \frac{1}{m}, \tag{23}$$

which implies $\min_r F_{\mathrm{e}}(\mathcal{C}_r, \mathrm{id}) \le 1/m$. Conversely, choosing $\mathcal{R}_a = \mathrm{id}$ gives $\mathcal{C}_r = \Delta$ for every r, attaining $1/m$. □

Theorem 4 (Exact frame-aware optimum)**.** *If the controller knows r and may use recovery maps $\mathcal{R}_{a,r}$, then*

$$\sup_{\{\mathcal{R}_{a,r}\}} F_{\mathrm{e}}\left(\frac{1}{m}\sum_a \mathcal{R}_{a,r} \circ \Lambda_{a,r}^{(\lambda)}, \mathrm{id}\right) = 1 - \lambda + \frac{\lambda}{m}. \tag{24}$$

The upper bound follows because the coherent component contributes at most one to the entanglement fidelity, while any CPTP postprocessing of the dephasing component remains entanglement breaking and contributes at most $1/m$. The bound is attained by

$$\mathcal{R}_{a,r} = \mathrm{Ad}_{Z^{-a-r}U^\dagger}\,. \tag{25}$$

Therefore the exact minimax value of the missing outcome-frame information is

$$\left(1 - \lambda + \frac{\lambda}{m}\right) - \frac{1}{m} = (1-\lambda)\left(1 - \frac{1}{m}\right). \tag{26}$$

The theorem concerns a single-round or pre-calibrated recovery rule whose characterization data are restricted to the induced POVM, the nonselective channel, and the reported local outcome. It does not preclude learning r through additional branch-resolved process tomography, a trusted global wiring map, or an adaptive multi-round identification protocol. Those procedures supply precisely the outcome-frame information excluded by the marginal-only model.

The earlier Fourier-uniform state example remains a useful explicit mismatch: for one shifted branch, the frame-aware and unshifted corrections differ in state fidelity by $1 - \lambda$. That example should not be confused with the optimal worst-sector separation above, whose exact gap is $(1-\lambda)(1-1/m)$.

7 Numerical red-team checks

An independent numerical script samples random CPTP recovery channels via Haar-random Stinespring isometries. For $m = 2, 3, 4$, $\lambda \in \{0, 0.2, 0.5, 0.9, 1\}$, and 500 random marginal-only recovery families per case (7500 families total), the audit found no sampled violation of the $1/m$ minimax upper bound. Identity recovery reached $1/m$ in every sector to floating-point precision, and the sector-aware inverse reached $1 - \lambda + \lambda/m$. The sector-independence of the two marginals was also verified numerically. These computations are falsification-oriented cross-checks, not substitutes for the analytic proofs.

8 Relation to prior work and scope

Channel and Kraus-frame holonomy are established [11], and environment-assisted recovery from outcome-resolved random-unitary noise is established [12, 13]. Quantum-instrument compatibility, interactive-instrument resource theory, and midcircuit-measurement characterization likewise make clear that outcome-conditioned structure can matter beyond averaged observables [6, 9, 10, 14, 15]. Accordingly, we do not claim that conditional branches matter for the first time, that average channels are universally sufficient, or that the cyclic unitary family defines new fundamental physics.

The narrower contribution is the combination of: fixed POVM and fixed nonselective channel; a nondegenerate outcome-frame covering and its monodromy; a global labeling obstruction; robust finite-sample branch matching; quantum branch certification; and an exact minimax limit on recovery rules restricted to the two marginals and the reported local outcome. The work remains entirely within standard finite-dimensional quantum mechanics. It does not establish hidden variables, a modification of the Born rule, or a new fundamental eight-state ontology.

Several limitations remain. The covering construction excludes branch-degeneracy loci. The finite-sample bound is deliberately conservative. The exact minimax theorem is for a specific noisy cyclic family and a specific information restriction, not a universal theorem for all instruments. A hardware demonstration would have to audit classical wiring, software remapping, drift, and SPAM errors separately so that an observed branch permutation is not imposed by the control stack.

9 Experimental route

The smallest illustrative implementation uses $m = 2$ and $q = 4$. An ancilla can generate a uniformly random recorded outcome, followed by an outcome-conditioned Z operation and a continuously varied diagonal phase. The protocol would: (i) verify identical POVMs and nonselective channels; (ii) reconstruct branch Choi states and the five edge assignment matrices including the closing edge; (iii) certify the exchange monodromy; (iv) verify branch Bell overlap above $1/2$; and (v) compare frame-aware and marginal-only feedback. This remains a proposed experiment, not an experimental result.

10 Data and code availability

The accompanying reproducibility archive deposited with this preprint contains the numerical audit of the cyclic family, the exact-minimax red-team script, machine-readable outputs, and a README with reproduction commands. The analytic claims are intended to be independently checkable from the manuscript and Supplemental Material.

Author contributions and use of AI-assisted tools

Y. Wang conceived the originating structural question, selected the research direction, and is solely responsible for the scientific claims. OpenAI ChatGPT (GPT-5.6 Sol, accessed 2026) was used under the author's direction for literature organization, mathematical drafting, code scaffolding and execution checks, proof-checking prompts, and document preparation. The included numerical outputs were regenerated before this archival release. No AI system is listed as an author.

Competing interests

The author declares no competing interests and no external funding for this work.

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